PROBLEM STATEMENT-

Multi-Input Digital Authorization Gate

Design a combinational logic circuit that unlocks a secure door only when a specific 3-bit binary code is entered. Any incorrect input should trigger an error signal.

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Components Used-

Made using TinkerCad

- **3 Slideswitches**
- **Inverter gate (NOT)**
- **Three input AND gate**
- **1 PowerSupply (V=5,Current=0.1 ohm)**
- **2 220 ohm resistor**
- **1 Green LED**
- **1 Red LED**

Secret code=110

Logic -

(` represents bar/complement)

You have:

- 3 input switches → represent a 3-bit binary code (A, B, C)
- One correct code (i.e 110)
- Two outputs:
  - 🟢 Green LED → correct code entered (door unlock)
  - 🔴 Red LED → wrong code entered (error)

<u>Unlock signal (Door Open)</u>

This should be HIGH only when:

A = 1, B = 1, C = 0

So we directly write:

Unlock= A.B. (C)`

<u>Error signal</u>

This should be HIGH for all other combinations except the correct one.

Error=(Unlock)`

Meaning=

Error = (A·B·(C)`)`

Using DeMorgan's Law-

Error = $(A)^{`} + (B)^{`} + C$

Truth Table-

Switch A	Switch B	Switch C	Unlock	Error
0	0	0	0	1
0	0	1	0	1
0	1	0	0	1
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	1	0
1	1	1	0	1

Working-

- **The circuit starts with three slide switches representing inputs A, B, and C. These switches act as a 3-bit binary code input. Each switch gives either logic 0 or 1 depending on its position.**

- **The input C is connected to a NOT gate. This is because the correct password is 110, so C must be 0. The NOT gate converts C into C` (C complement).**

Now, the outputs A, B, and C` are given to a 3-input AND gate. The AND gate checks whether all inputs are 1 simultaneously.

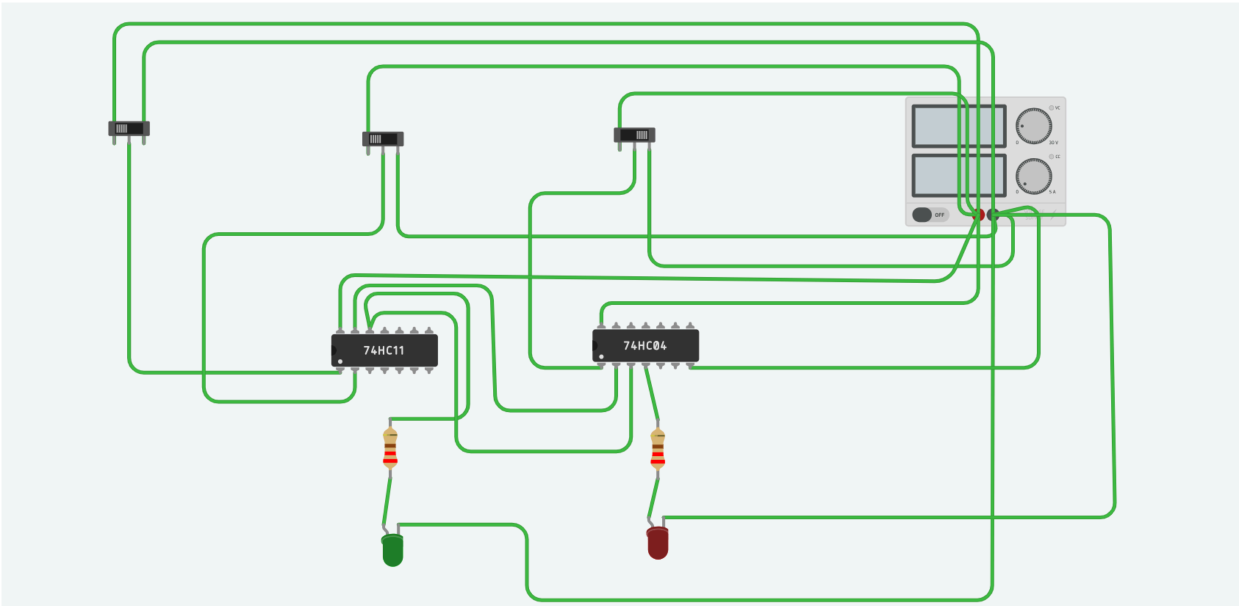
So:

- **A = 1**
- **B = 1**
- **C` = 1 (means C = 0)**

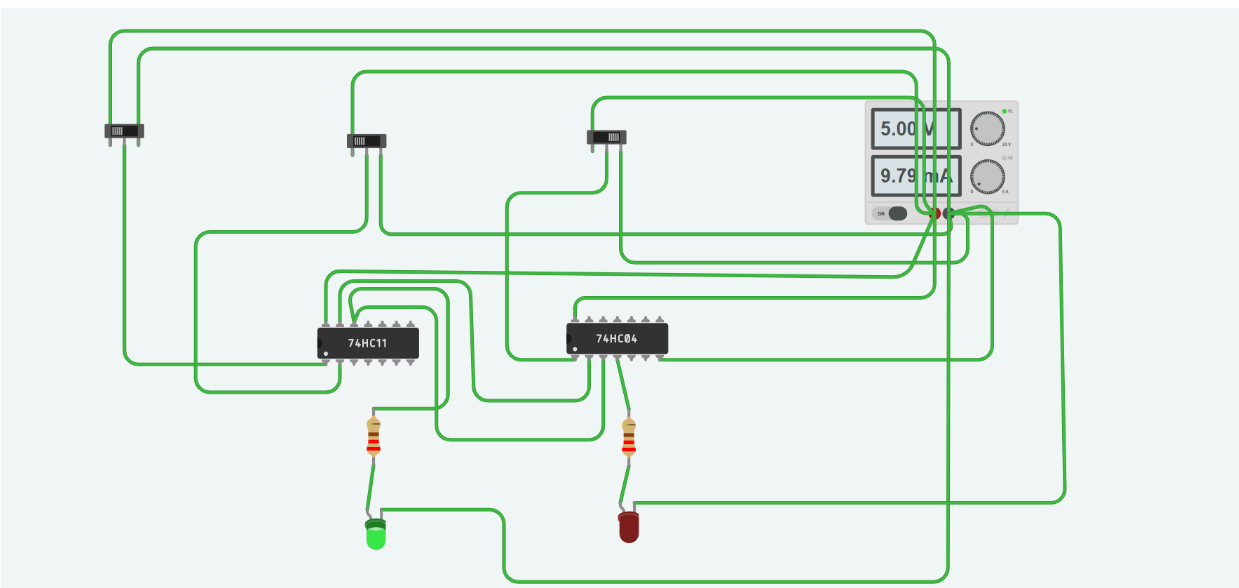
If all these conditions are satisfied, the AND gate output becomes HIGH.

- **The output of the AND gate is connected to the green LED through a resistor. So when the correct code 110 is entered, the AND output becomes 1 and the green LED glows, indicating successful unlock**
- **The same Unlock output is also connected to a NOT gate. This NOT gate generates the complement signal, which acts as the Error signal.**
- **So whenever the correct code is NOT entered, the Unlock is 0 and the NOT gate output becomes 1, which turns ON the red LED indicating wrong input.**
- **All switches and LEDs are powered using a 5V supply. Each LED is connected with a 220Ω resistor to limit current and protect the components. All grounds are connected together to complete the circuit.**

CIRCUIT DIAGRAM

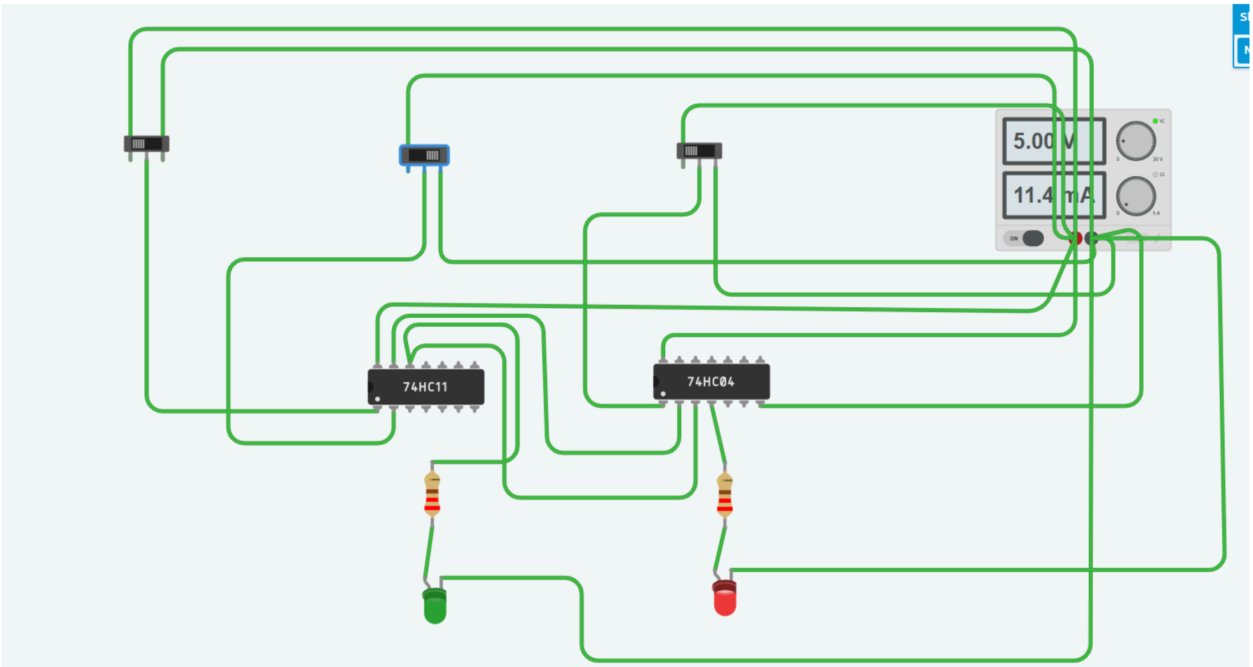


When Switch A= HIGH, SWITCH B=HIGH, SWITCH C=LOW



We can see the LED Light (green) is glowing as we enter the correct code.

And When the INPUT is any other expect this



You can see red LED Light is glowing.

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